习题课5

- 第八-十章作业讲评
- 一阶电路的时域分析

正弦稳态电路复习

1、复阻抗和复导纳

$$Z = \frac{\dot{U}}{\dot{I}} = |Z| \angle \varphi = R + jX$$

$$\begin{cases} |Z| = \dfrac{U}{I} & \text{阻抗模} \\ \varphi_Z = \varphi_u - \varphi_i & \text{阻抗角} \end{cases}$$

$$Y = \frac{\dot{I}}{\dot{U}} = G + jB = |Y| \angle \varphi'$$

$$\begin{cases} |Y| = \dfrac{I}{U} \\ \varphi_Y = \varphi_i - \varphi_u \end{cases}$$

$$Z = \frac{1}{Y}, Y = \frac{1}{Z}$$

$$\begin{cases} |Z||Y| = 1 \\ \varphi_Z + \varphi_Y = 0 \end{cases}$$

2023年6月3日

雨课堂 Rain Classroom

2、阻抗（导纳）的串联和并联

和直流电阻电路分析方法一样，有分压和分流公式。

3、相量图

4、正弦稳态电路的分析

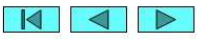
①先画相量运算电路 { 电压、电流→相量；复阻抗 }

②相量形式KCL、KVL定律，欧姆定律

③直流电阻电路的定理和分析方法都适用

④相量图

2023年6月3日



5、正弦稳态电路的功率

有功（平均）功率：$P = UI\cos\varphi$

无功功率：$Q = UI\sin\varphi$

视在功率：$S = UI$

复功率：$\overline{S} = \dot{U}\,\dot{I}^{*} = P + \mathrm{j}Q$

$$P = UI\cos\varphi = I^2|Z|\cos\varphi = I^2R = \frac{Q}{\tan\varphi} = \mathrm{Re}[\overline{S}]$$

$$Q = UI\sin\varphi = I^2|Z|\sin\varphi = I^2X = P\tan\varphi = \mathrm{Im}[\overline{S}]$$

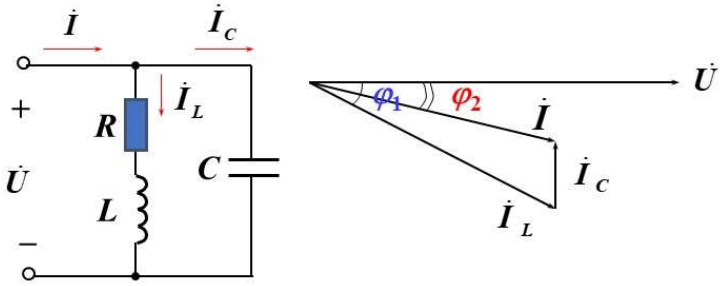
$$S = |\overline{S}| = \sqrt{P^2 + Q^2}$$

$$\overline{S} = \dot{U}\,\dot{I}^{*} = \dot{I}\,\dot{I}^{*}Z = I^2Z = U^2Y^{*}$$

2023年6月3日



6、功率因数的补偿



并联电容前　$\dot{I} = \dot{I}_L$

并联电容后　$\dot{I} = \dot{I}_L + \dot{I}_C$

并联前后　$\dot{U}$ 不变，$\dot{I}_L$ 也不变，但　$\dot{I}$ 减小了。

$$C = \frac{P}{\omega U^2}(\tan\ \varphi_1 - \tan\ \varphi_2)$$

7、最大功率传输

$$Z_L = Z_{eq}^* \qquad P_{\max} = \frac{U_{OC}^2}{4R_{eq}}$$

2023年6月3日

8、串联电路谐振

谐振条件：　$\mathrm{Im}[\ Z(j\omega)] = 0$

谐振频率：　$\omega_0 = \dfrac{1}{\sqrt{LC}}$

$$Q = \frac{U_L(\omega_0)}{U} = \frac{U_C(\omega_0)}{U} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

谐振时，端口的电压和电流同相，阻抗模最小，电流最大，平均功率最大。

当 $Q >> 1$时，$U_L = U_C >> U$，出现过电压现象。

2023年6月3日

9、并联电路谐振

谐振条件：　$\mathrm{Im}[\ Y(j\omega)] = 0$

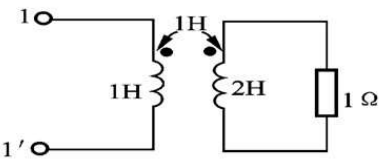
谐振频率：　$\omega_0 = \dfrac{1}{\sqrt{LC}}$

$$Q = \frac{\omega_0 C}{G} = \frac{1}{\omega_0 LG} = \frac{1}{G}\sqrt{\frac{C}{L}}$$

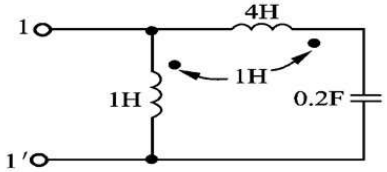
谐振时，端口的电压和电流同相，导纳模最小，阻抗模最大，电压最大，平均功率最大。

2023年6月3日

8-1、试求图所示电路的输入阻抗Z(w=1rad/s)。



(a)



(b)

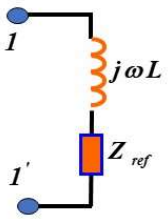
解

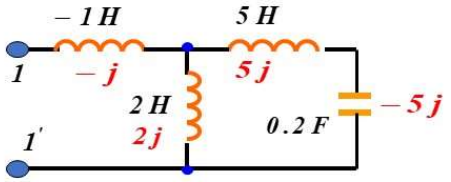
$$Z_{ref} = \frac{(\omega M)^2}{Z_{22}} = \frac{1}{1+j2}$$

$$Z_{11'} = j\omega L_1 + Z_{ref}$$

$$= j + \frac{1}{1+j2}$$

$$= j + \frac{1-j2}{5} = \frac{1}{5}(1+j3)$$

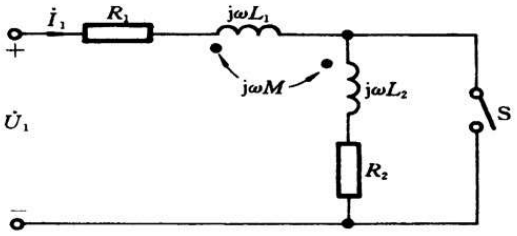




$$Z_{11'} = -j$$

8-2 如图所示电路中，$R_1=R_2=1\Omega$，$\omega L_1=\omega L_2=2\Omega$，$\omega M=1\Omega$，$U_1=100V$。求：

（1）开关S打开和闭合时的电流 $\dot{I}_1$　　（2）S闭合时各部分的复功率。



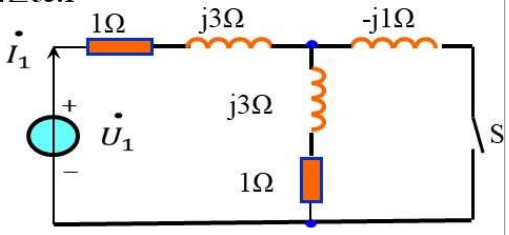
解法一　（1）开关S打开时

$$Z = 2 + j6$$

$$\dot{I}_1 = \frac{\dot{U}_1}{Z} = \frac{100}{2+j6} = 15.8\angle -71.6^\circ \text{ A}$$

（2）开关S闭合时

$$Z = (1+j3) + (1+j3)//(-j1) = \frac{6+j8}{5} = 2\angle 53.1^\circ$$



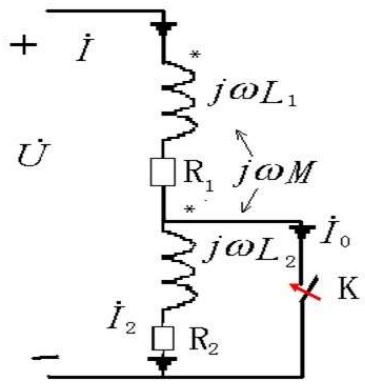
$$\dot{I}_1 = \frac{\dot{U}_1}{Z} = 50\angle -53.1^\circ \text{ A}$$

复功率　$S = \dot{U}_1 \dot{I}_1^*$ $=100\times 50\angle 53.1^\circ$

$=3000+j4000(VA)$

线圈1复功率 $\overline{S_1}=\overline{S}$，线圈2复功率 $\overline{S_2}=0$

解法二：



解: (1) K打开时

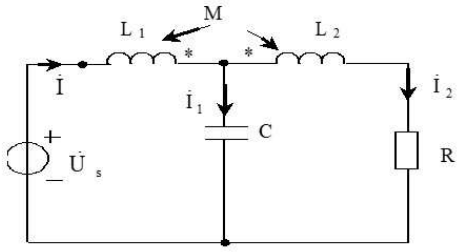
$$\dot{I} = \frac{\dot{U}}{R_1 + R_2 + j\omega(L_1 + L_2 + 2M)}$$

(2) K闭合时

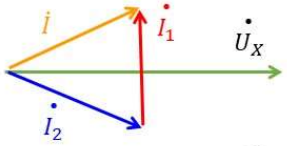
$$(R_1 + j\omega L_1)\dot{I} + j\omega M \dot{I}_2 = \dot{U}$$

$$j\omega M \dot{I} + (R_2 + j\omega L_2)\dot{I}_2 = 0$$

8-3 图 示正弦交流电路中，已知$I_1=I_2=I=10A$，$X_C=-10\Omega$，电路吸收有功功率433W。求R，X_{L2}及ωM。



解　去耦、作相量图，等边△



注意$\dot{I}_1$与$\dot{U}_X$的相位关系

电阻参数　$R=\dfrac{P}{I_2^2}=4.33\Omega$

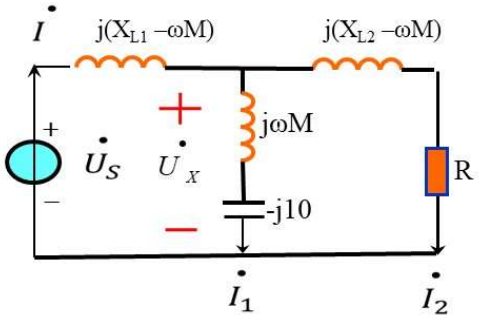
感性阻抗角30度　$\dfrac{R}{X_{L2}-\omega M}=\sqrt{3}$

阻抗模相等

$(X_{L2}-\omega M)^2+R^2=(\omega M-10)^2$

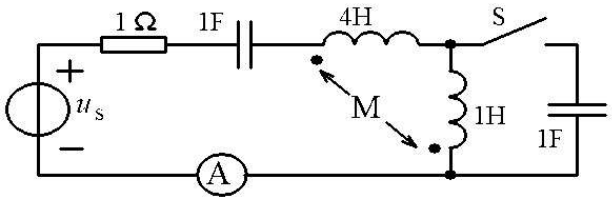
$\omega M=5\Omega$

$X_{L2}=7.5\Omega$



8-4、如图所示含耦合互感的电路。S打开时，电流表读数3A；S闭合时，电流表读数为0。已知电源 $u_s=3\sqrt{2}\cos(\omega t+4^\circ)$

求电源的角频率ω和互感M的参数。



解　S打开时，电路发生串联谐振

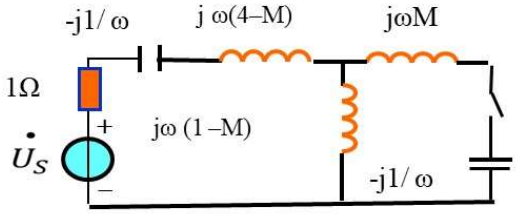
$j[\omega(4-M)+\omega(1-M)-1/\omega]=0$

S闭合时，电路发生并联谐振

$j\omega(1-M)+j(\omega M-1/\omega)=0$

$\omega=1rad/s$

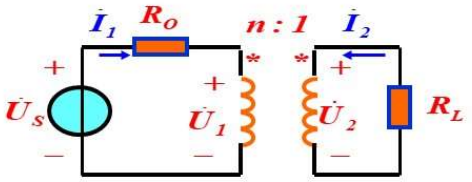
$M=2H$



8-5、在如图所示电路中，已知 $\dot{U}_s = 100\angle 30^\circ$ $R_O = 100\,\Omega$, $R_L = 1\,\Omega$ 匝比n=5。求：电流 $\dot{I}_1$和$\dot{I}_2$

解

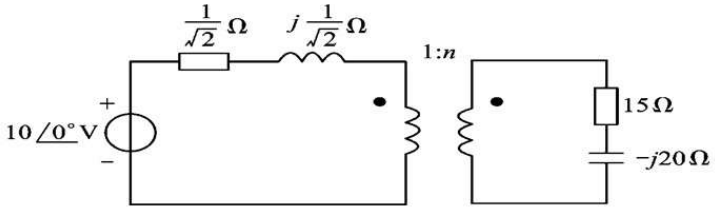
$$\begin{cases} \dot{U}_s = R_O \dot{I}_1 + \dot{U}_1 \\ \dot{U}_2 = -\dot{I}_2 R_L \\ \dot{U}_1 = n\dot{U}_2 \\ \dot{I}_1 = -\dfrac{1}{n}\dot{I}_2 \end{cases}$$



$$\dot{I}_1 = \frac{4}{5}\angle 30^\circ \qquad \dot{I}_2 = 4\angle(-150^\circ)$$

8-6 电路如图所示，为使负载Z_L=15-j20能获得最大功率，试确定理想变压器的变比*n*, 并求最大功率值。

解 **阻抗模值匹配,有最大功率**



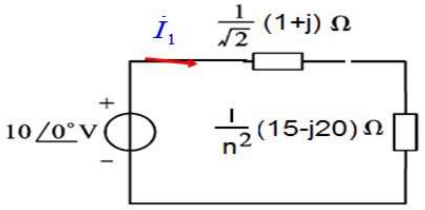
$$\frac{\sqrt{1^2+1^2}}{\sqrt{2}} = \frac{\sqrt{15^2+(-20)^2}}{n^2}$$

$$n = 5$$

$$Z = \frac{1}{\sqrt{2}}(1+j) + \frac{1}{5^2}(15 - j20) = 1.31\angle -4.07^\circ$$

$$\dot{I}_1 = \frac{\dot{U}}{Z} = \frac{10\angle 0^\circ}{1.31\angle -4.07^\circ} = 7.63\angle 4.07^\circ$$

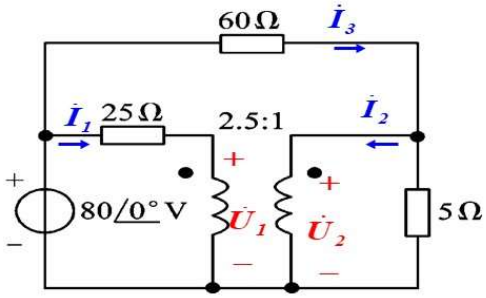
$$P = (\frac{1}{n}I_1)^2 \times 15 = (\frac{7.63}{5})^2 \times 15 = 35\,W$$



8-7、 电路如图所示，求每个电阻所消耗的功率。

解

$$\begin{cases} 80\angle 0^\circ = 25\,\dot{I}_1 + \dot{U}_1 \\ \dot{U}_2 = 5\times(\dot{I}_3 - \dot{I}_2) \\ 80\angle 0^\circ = 60\,\dot{I}_3 + \dot{U}_2 \\ \dot{U}_1 = 2.5\dot{U}_2 \\ \dot{I}_1 = -\dfrac{1}{2.5}\dot{I}_2 \end{cases}$$



$$\rightarrow \begin{cases} \dot{I}_1 = 1.2\angle 0^\circ \\ \dot{U}_2 = 20\angle 0^\circ \\ \dot{I}_3 = 1\angle 0^\circ \end{cases}$$

$$P_{25} = 25\,I_1^2 = 36\,W$$

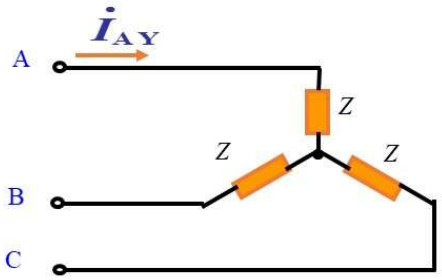
$$P_5 = U_2^2/5 = 80\,W$$

$$P_{60} = 60\,I_3^2 = 60\,W$$

9-1 对称三相电路的线电压U1=230V，负载阻抗
Z=（12+j16）Ω。试求：
（1）星形连接负载时的线电流及吸收的总功率；
（2）三角形连接负载时的线电流、相电流和吸收的总功率；
（3）比较（1）和（2）的结果能得到什么结论？

解 （1）

$$Z = 12 + j16 = 20\angle 53.1^\circ$$



$$U_l = 230$$

$$I_{lY} = \frac{U_P}{|Z|} = \frac{U_l}{\sqrt{3}}\cdot\frac{1}{20} = 6.64A$$

$$P_Y = \sqrt{3}U_l I_{lY}\cos\varphi_Z$$

$$= \sqrt{3}\cdot 230\cdot\frac{230}{\sqrt{3}\cdot 20}\cos 53.1^\circ = 1587W$$

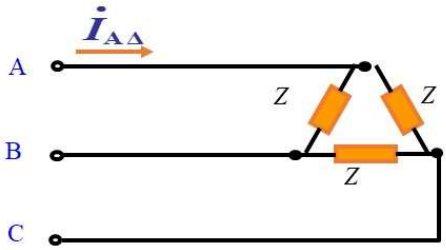
（2）三角形连接负载时的线电流、相电流和吸收的总功率；

$$Z = 12 + j16 = 20\angle 53.1^\circ$$

$$U_l = 230$$

$$I_p = \frac{U_l}{|Z|} = \frac{U_l}{20} = 11.5A$$

$$I_{l\Delta} = \sqrt{3} I_P = 19.9A$$



$$P_\Delta = \sqrt{3} U_l I_{l\Delta} \cos\varphi_Z = \sqrt{3} \times 230 \times 19.9 \times 0.6 = 4761W$$

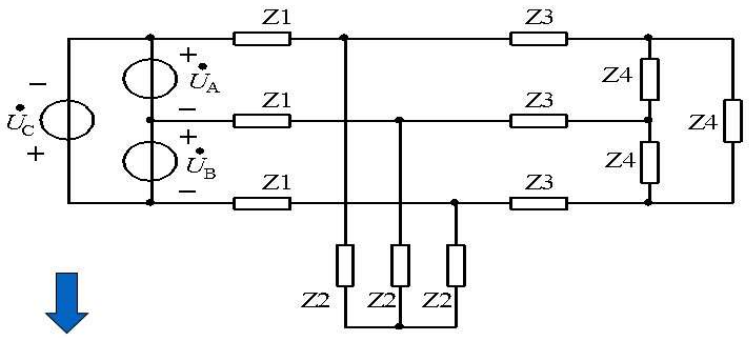
（3）比较（1）和（2）的结果能得到什么结论？

$I_{l\Delta} = 3I_{lY}$　　$P_\Delta = 3P_Y$

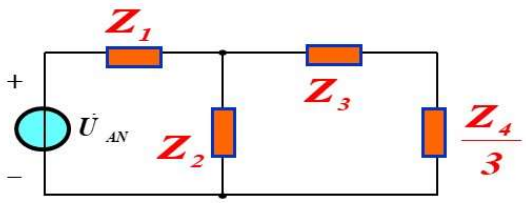
结论：在同样的线电压时，Y接法线电流小，吸收的总功率小；△接法线电流大，吸收的总功率大。

9-2 图所示对称三相电路。用三相归结为一相的方法分析计算。请画出一相计算电路（A相），并在图中标出必要的参数。

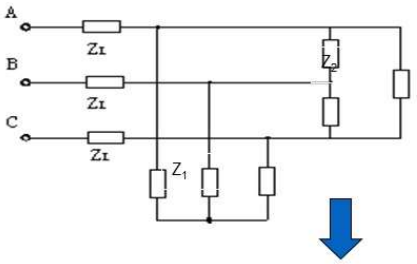
解



$$\dot{U}_{AN} = \frac{1}{\sqrt{3}} \angle -30^\circ \dot{U}_A$$



9-3、 图示电路，三相电源线电压**380V**，对两组对称负载供电。**Y**联接负载每相复阻抗Z_1=12+*j*16Ω，Δ联接负载每相复阻抗Z_2=48+*j*36Ω，每根传输导线的复阻抗Z_L=1+*j*2Ω。试求流过端线阻抗的电流及三相电源的功率。



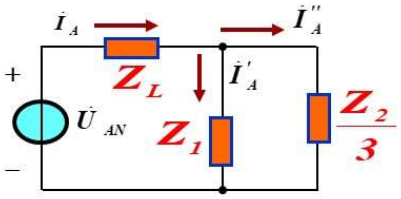
解 用三相归结为一相的方法分析计算。

设：$\dot{U}_{AB}=380\angle 30^\circ$ V　则：$\dot{U}_{AN}=220\angle 0^\circ$ V

$$Z=Z_L+Z_1//\frac{Z_2}{3}=1+j2+\frac{(12+j16)(16+j12)}{12+j16+16+j12}$$

$$=8.14+j9.14=12.24\angle 48.3^\circ$$

$$\dot{I}_A=\frac{\dot{U}_{AN}}{Z}=\frac{220\angle 0^\circ}{12.24\angle 48.3^\circ}\approx 18\angle(-48.3^\circ)\text{A}$$

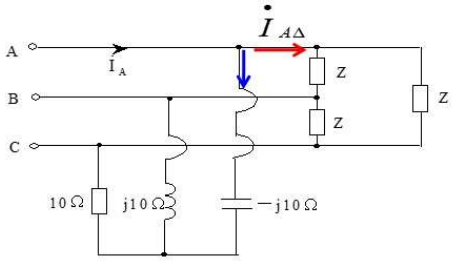


$$\text{P}=\sqrt{3}U_l I_l\cos\varphi$$

$$=\sqrt{3}\times 380\times 18\times\cos 48.3^\circ=7878.2W$$

9-4 图为对称三相电源，线电压 $\dot{U}_{AB}=100\sqrt{3}$ V，阻抗 $Z=30+j30$ Ω

求 $\dot{I}_A$



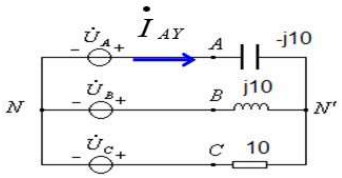
解 设：$\dot{U}_{AB}=100\sqrt{3}\angle 30^\circ$ V　则：$\dot{U}_{AN}=100\angle 0^\circ$ V

（1）对称△三相电路

$$\dot{I}_{A\Delta}=\sqrt{3}\angle -30^\circ\frac{\dot{U}_{AB}}{Z}=5\sqrt{2}\angle -45^\circ$$

（2）不对称Y三相电路

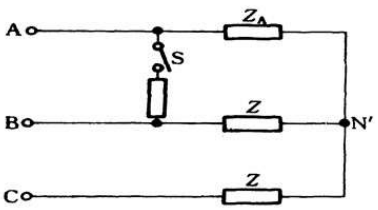
$$\left(\frac{1}{-j10}+\frac{1}{j10}+\frac{1}{10}\right)\dot{U}_{N'}=\frac{100\angle 0^\circ}{-j10}+\frac{100\angle -120^\circ}{j10}+\frac{100\angle 120^\circ}{10}$$



$$\dot{I}_{AY}=\frac{100\angle 0^\circ-\dot{U}_{N'}}{-j10}\qquad\Rightarrow\dot{I}_{AY}=(15\sqrt{2}+5\sqrt{6})\angle 45^\circ$$

$$\dot{I}_A=\dot{I}_{AY}+\dot{I}_{AY}=(20+5\sqrt{3})+j(10+5\sqrt{3})=34.2\angle 33^\circ A$$

9-5 如图所示电路中，对称三相电源端的线电压U_1=380V，Z=（50+j50）Ω，Z_1=（100+j100）Ω，Z_A为R、L、C串联组成，R=50Ω，X_L=314Ω，X_C=-264Ω。试求：
（1）开关S打开时的线电流；
（2）若用二瓦计法测量电源端三相功率，试画出接线圈，并求两个功率表的读数（S闭合时）。



解

设：$\dot{U}_{AB} = 380\angle 30^\circ \Rightarrow \dot{U}_{AN} = 220\angle 0^\circ$

$$Z_A = 50 + j314 - j264 = 50 + j50(\Omega) = Z$$

（1）开关S打开时,对称Y三相电路

$$\dot{I}_A = \frac{\dot{U}_{AN}}{Z} = \frac{220\angle 0^\circ}{50 + j50} = 3.11\angle -45^\circ \Rightarrow I_l = 3.11A$$

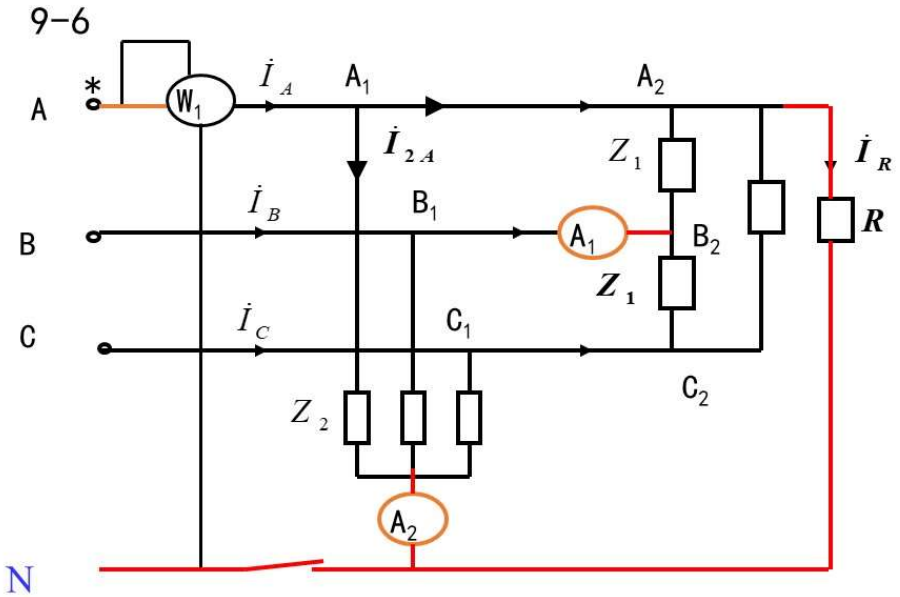
（2）两个功率表如图接线

$$\dot{I}_{AB} = \frac{\dot{U}_{AB}}{Z_1} = \frac{380\angle 30^\circ}{100 + j100} = 2.687\angle -15^\circ$$

$$\dot{I}_A = 3.11\angle -45^\circ + 2.687\angle -15^\circ \qquad \dot{I}_B = 3.11\angle -165^\circ - 2.687\angle -15^\circ$$

$$P_1 = \mathrm{Re}(\dot{U}_{AC} \times \dot{I}_A^*) = 2128\ W \qquad P_2 = \mathrm{Re}(\dot{U}_{BC} \times \dot{I}_B^*) = 42\ W$$

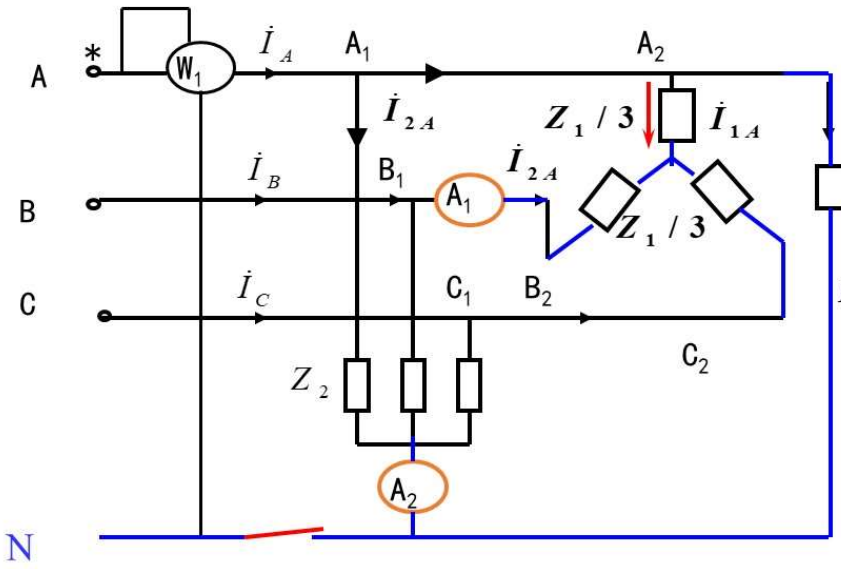
9-6



(1) 开关闭合时各表的读数;

(2) 开关打开时各表的读数.

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(1) 开关闭合时

$$\dot{I}_{2A} = \frac{\dot{U}_A}{Z_2} = \frac{220\angle 0^\circ}{13\angle 67.4^\circ} A$$

$$\dot{I}_{1A} = \frac{\dot{U}_A}{Z_1/3} = 65.8\angle 90^\circ A$$

$$R = \frac{U_A^2}{P_R} = \frac{220^2}{24200} = 2\Omega$$

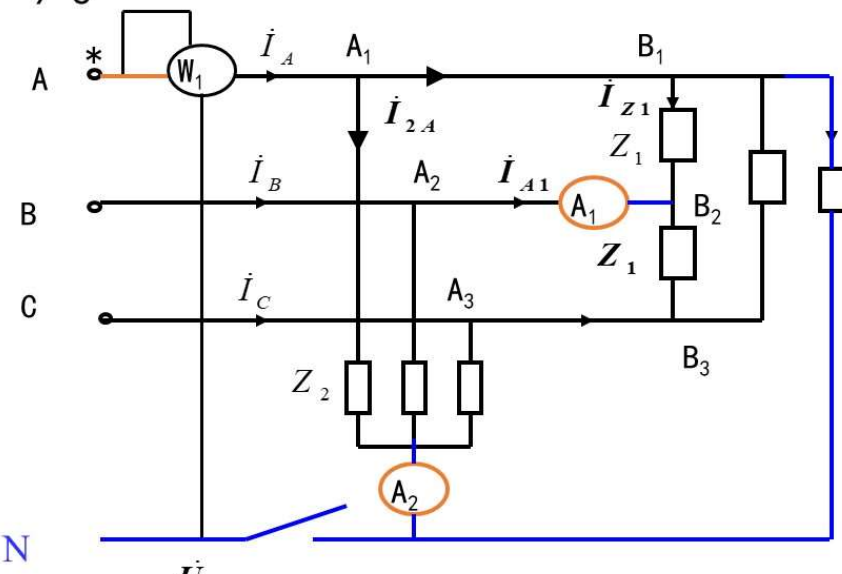
$$\dot{I}_A = \dot{I}_{2A} + \dot{I}_{1A} + \dot{I}_R = 126.88\angle 23.4^\circ A$$

$$P_A = U_A I_A \cos 23.4^\circ = 25680 \quad w$$

$$P = (P_A - P_R)\times 2 + P_A$$



9-6



(2) 开关打开时

$$I_{Z1} = \frac{U_{AB}}{|Z_1|} = 38\ A$$

$$\dot{I}_{A1} = 1.732 \times I_{Z1} = 65.8\ A$$

断开R,求戴氏等效电路(与Z1无关)

$$\dot{U}_{oc} = \dot{U}_{A0} = 220\angle 0$$

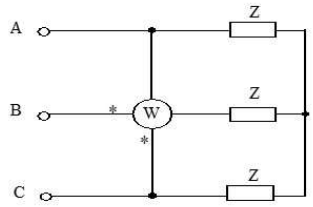
$$Z_{eq} = \frac{1}{3}Z_2 = 5/3 + j4$$

$$\dot{I}_R = \frac{\dot{U}_{oc}}{R + Z_{eq}} = 40.5\angle 47.46^\circ$$

功率表的读数无意义 .



9-7 图所示对称三相电路中，负载消耗功率为**3kW**，电路功率因数为**0.866**(负载为感性)，试求功率表的读数。



解　　设：$\dot{U}_{AB}=U_l\angle 0^\circ\ \mathrm{V}$

$\dot{U}_{BC}=U_l\angle(-120^\circ)\mathrm{V}$，$\dot{U}_{CA}=U_l\angle 120^\circ$，$\dot{U}_{BN}=\dfrac{U_l}{\sqrt{3}}\angle -150^\circ\ \mathrm{V}$

$$\dot{I}_B=\frac{\dot{U}_{BN}}{Z}=\frac{\frac{U_l}{\sqrt{3}}\angle(-150^\circ)}{|Z|\angle\varphi}=I_l\angle -180^\circ$$

$\because \cos\varphi=0.866$

$\therefore \varphi=30^\circ$

$$P=\sqrt{3}U_lI_1\cos\varphi \qquad \therefore U_lI_1=\frac{P}{\cos\varphi}=\frac{3000}{\sqrt{3}\times 0.866}=2000$$

$$P_W=|\dot{U}_{CA}|\cdot|\dot{I}_B|\cos\varphi_{(\dot{U}_{CA}-\dot{I}_B)}$$

$$\varphi_{(\dot{U}_{CA}-\dot{I}_B)}=\psi_{\dot{U}_{CA}}-\psi_{\dot{I}_B}=120^\circ+180^\circ=-60^\circ$$

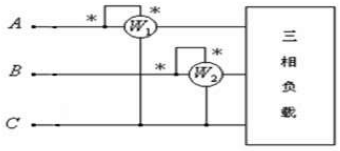
$$\therefore \mathrm{P_W}=U_lI_1\cos(-60^\circ)=1000W$$

9-8 对称三相电路的负载吸收的功率为**2.4kW**，功率因数为**0.4**（感性）。

试求：（**1**）两个功率表的读数（用二瓦计法测量功率时）；

（**2**）怎样才能使负载端的功率因数提高到**0.8**？再求出两个功率表的读数。

解　　(1) 两个功率表如图接线



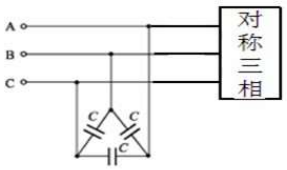
$$P=\sqrt{3}U_lI_1\cos\varphi=2400$$

$$\Rightarrow U_lI_1=2000\sqrt{3},\varphi=66.4^\circ$$

W1 $=U_lI_1\cos(\varphi-30)^0=2787.5W$

W2 $=U_lI_1\cos(\varphi+30)^0=-387.5W$

(2) 如图接对称△或Y电容网络，提高功率因数



$$P=\sqrt{3}U_lI_l'\cos\varphi'=2400$$

$$\Rightarrow U_lI_l'=1000\sqrt{3},\varphi'=36.9^\circ$$

W1 $=U_lI_l'\cos(\varphi'-30)^0=1719.7W$

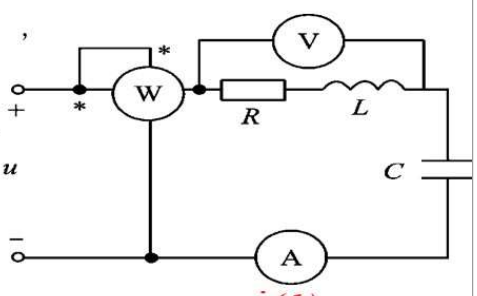
W2 $=U_lI_l'\cos(\varphi'+30)^0=680.3W$

10-1在图所示的电路中，已知 $R=10\Omega$

$\omega L=10\Omega$ $1/\omega C=90\Omega$

$u(t)=[100+150\cos\omega t+100\cos(2\omega t-90°)]V$

求图示电路中各仪表的读数。



解 (1)直流分量作用：

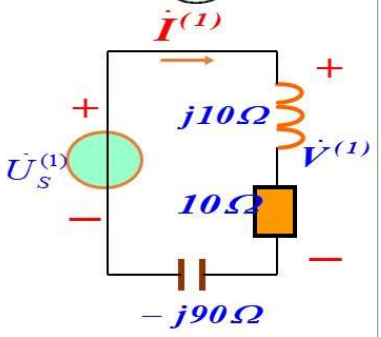
$$I^{(0)}=0 \qquad U^{(0)}=0 \qquad P^{(0)}=0$$

(2)基波分量作用：

$$\dot{I}^{(1)}=\frac{\dot{U}_S^{(1)}}{10+(j10-j90)}=\frac{150/\sqrt{2}}{10-j80}=\frac{1.86}{\sqrt{2}}\angle 82.9°$$

$$\dot{U}^{(1)}=(10+j10)\dot{I}^{(1)}=18.6\angle 127.9°$$

$$P^{(1)}=(I^{(1)})^2R=(\frac{1.86}{\sqrt{2}})^2\cdot 10=17.3W$$



$\omega L=10\Omega$ $1/\omega C=90\Omega$ $R=10\Omega$

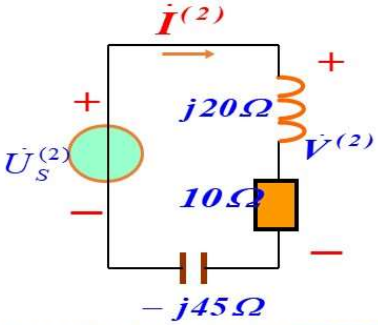
$u(t)=[100+150\cos\omega t+100\cos(2\omega t-90°)]V$

(3) 2次谐波作用：

$$\dot{I}^{(2)}=\frac{\dot{U}_S^{(2)}}{10+(j20-j45)}=\frac{100/\sqrt{2}}{10-j25}=\frac{3.71}{\sqrt{2}}\angle 68.2°$$

$$\dot{U}^{(2)}=(10+j20)\dot{I}^{(2)}=\frac{82.95}{\sqrt{2}}\angle 131.63°$$

$$P^{(2)}=(I^{(2)})^2R=(\frac{3.71}{\sqrt{2}})^2\cdot 10=68.8W$$



$$I=\sqrt{(I^{(0)})^2+(I^{(1)})^2+(I^{(2)})^2}=\sqrt{(\frac{1.86}{\sqrt{2}})^2+(\frac{3.71}{\sqrt{2}})^2}\approx 2.93A$$

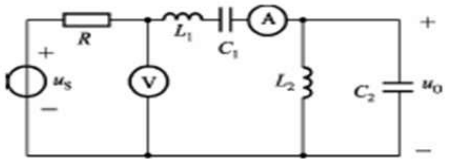
$$V=\sqrt{(U^{(0)})^2+(U^{(1)})^2+(U^{(2)})^2}=\sqrt{(\frac{18.6}{\sqrt{2}})^2+(\frac{82.95}{\sqrt{2}})^2}\approx 60.1V$$

$$P=P^{(0)}+P^{(1)}+P^{(2)}=17.3+68.8=86.1W$$

10-2 图示电路中，已知 $f = 500Hz$ $C_1 = C_2 = 3.18\mu F$, $R = 10\Omega$

$$u_s = 141\cos\omega t + 14.1\cos(3\omega t + 30°)\,V$$

电压表内阻视为无限大，电流表内阻视为零。已知基波电压单独作用时电流表读数为零；三次谐波电压单独作用时，电压表读数为零。求电感 L_1 L_2 和电容 C_2 两端的电压 u_0

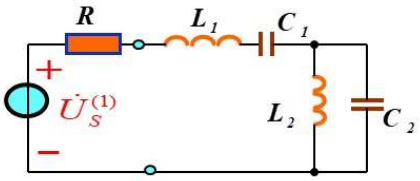


解 (1)基波分量单独作用时：

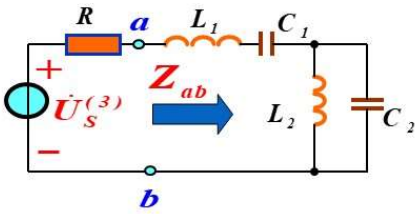
电路并联谐振 $\omega L_2 = \dfrac{1}{\omega C_2}$

$$L_2 = \frac{1}{\omega^2 C_2} = \frac{1}{(2\pi f)^2 3.18\cdot 10^{-6}} = 31.9\,mH$$

$$u_{c2}^{(1)} = u_s^{(1)} = 141\cos\omega t = 141\cos 3142\,t$$



(2)三次谐波电压单独作用时：



电路串联谐振 $u_{ab}^{(3)}(t) = 0$

$$Z_{ab} = j3\omega L_1 + \frac{1}{j3\omega C_1} + j3\omega L_2 \,//\, \frac{1}{j3\omega C_2} = 0$$

$$L_1 = \frac{1}{j3\omega}\left(-\frac{1}{j3\omega C_1} - j3\omega L_2 \,//\, \frac{1}{j3\omega C_2}\right) = 7.5\,mH$$

$$\dot{I}_R = \frac{\dot{U}_S^{(3)}}{R} = \frac{10\angle 30°}{10} = 1\angle 30°$$

$$Z_{L_2C_2} = j3\omega L_2 \,//\, \frac{1}{j3\omega C_2} = j70.9\Omega$$

$$\dot{U}_{C2}^{(3)} = \dot{I}_R Z_{L_2C_2} = 70.9\angle -60°$$

$$\Rightarrow\quad u_{c2}^{(3)} = 70.9\sqrt{2}\cos(9425\,t - 60°)$$

(3)**C2**电压：

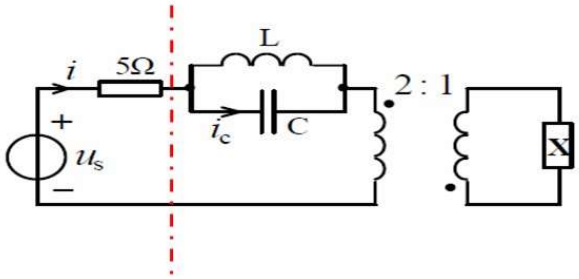
$$u_{c2} = u_{c2}^{(1)} + u_{c2}^{(3)} = 141\cos 3142\,t + 70.9\sqrt{2}\cos(9425\,t - 60°)$$

10-3 图示含有理想变压器的电路，已知电源 $u_s = \sqrt{2}\cos t + 15\sin 2t\ (V)$ ，电流 i = 3sin2t（A）,电感L=1H。试求：

（1）电容C 的参数值；

（2）X 应是什么元件？求出其参数；

（3）电流ic (t) 的有效值。



解 （1）对基波，LC并联谐振

$$\omega L = \frac{1}{\omega C}$$

$$C = \frac{1}{\omega^2 L} = \frac{1}{1^2 \times 1} = 1F$$

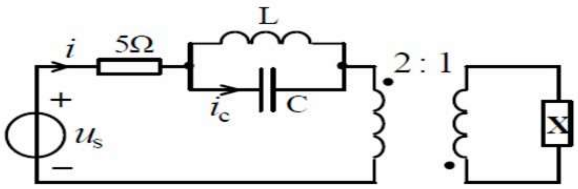
$$\dot{U}_C^{(1)} = 1\angle 0^\circ \qquad \dot{I}_C^{(1)} = j\omega C \dot{U}_C^{(1)} = j \times 1 \times 1 \times 1\angle 0^\circ = 1\angle 90^\circ$$

（2）对2次谐波，虚线右侧串联谐振

$$\frac{j2L \times (\frac{1}{j2C})}{j2L + \frac{1}{j2C}} + n^2 Z_X = 0 \qquad \Rightarrow Z_X = j\frac{1}{6}\Omega, L_X = \frac{1}{12}H$$

$$\because i = 3\sin 2t\ A \qquad \therefore \dot{I}^{(2)} = \frac{3}{\sqrt{2}}\angle -90^\circ$$

$$\dot{I}_C^{(2)} = \frac{j2L}{j2L + \frac{1}{j2C}} \times \dot{I}^{(2)} = 2\sqrt{2}\angle -90^\circ$$



（3）电流ic (t) 的有效值

$$I_C = \sqrt{(I_C^{(1)})^2 + (I_C^{(2)})^2} = \sqrt{1^2 + (2\sqrt{2})^2} = 3\ A$$

过渡过程1

重点：

1. 三要素法求一阶电路的阶跃响应和正弦激励下的响应.

$$f(t)=f(\infty)+[f(0^+)-f(\infty)\big|_{0^+}]e^{-\frac{t}{\tau}}$$

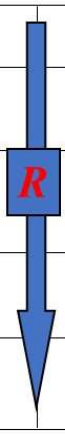
2. 分析动态电路响应的两种方法：

全响应＝零输入响应+零状态响应

全响应＝ 自由响应+强制响应

过渡过程2

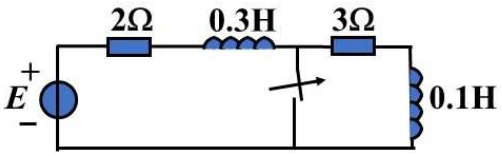
- 二阶电路的四种响应形式



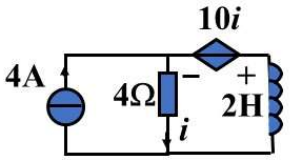
特征根	响应性质	自由分量形式
共轭虚根	等幅振荡 （无阻尼）	$K\sin(\omega_0 t+\theta)$
共轭复根	衰减振荡 （欠阻尼）	$Ke^{-\alpha t}\sin(\omega_d t+\theta)$ 或$e^{-\alpha t}(A\sin\omega_d t+B\cos\omega_d t)$
相等的实根	非振荡放电 （临界阻尼）	$e^{-\alpha t}(A_1+A_2 t)$
不等的实根	非振荡放电 （过阻尼）	$A_1e^{p_1 t}+A_2e^{p_2 t}$

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1. 求时间常数.



$\tau=(0.1+0.3)/5=0.08s$



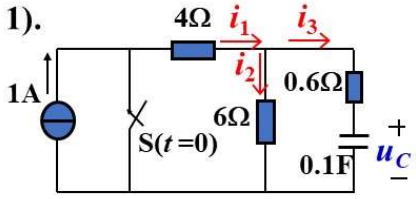
$u=10i+4i=14i$

$R_{等}=14\Omega$

$\tau=2/R_{等}=2/14=0.143s$

2. 求电压或电流的初始值。

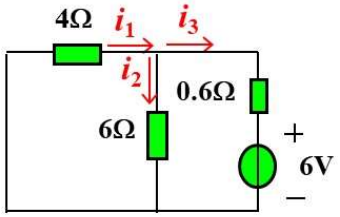
1).



求: $i_1(0^+), i_2(0^+), i_3(0^+)$.

解:

$t=0^+$时刻电路:



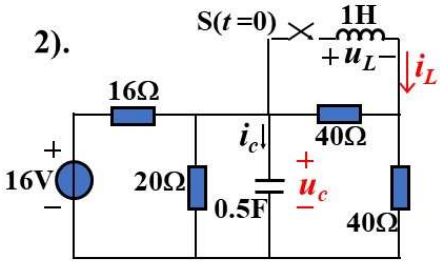
$$u_C(0^+)=u_C(0^-)=6\times1=6\text{ V}$$

$$i_3(0^+)=-\frac{6}{0.6+4//6}=-2\text{ A}$$

$$i_1(0^+)=\frac{i_3(0^+)\times6}{4+6}=-1.2\text{ A}$$

$$i_2(0^+)=-\frac{i_3(0^+)\times4}{4+6}=0.8\text{ A}$$

2).

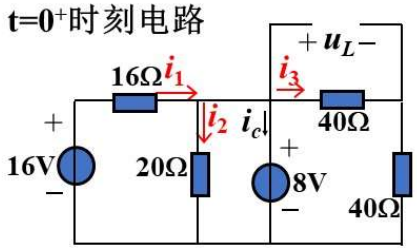


求：$i_C(0^+)$，$u_L(0^+)$ 和 $\left.\frac{di_L}{dt}\right|_{t=0^+}$

解：$i_L(0^+)=i_L(0^-)=0$

$$u_C(0^+)=u_C(0^-)=\frac{16\times(20//80)}{16+(20//80)}=8\ \text{V}$$

$t=0^+$时刻电路



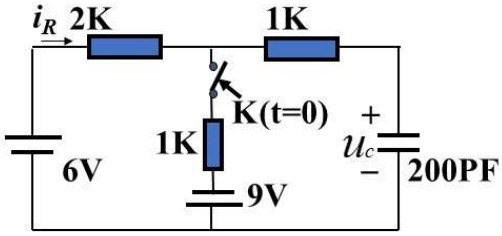
$$i_C(0^+)=i_1-i_2-i_3=\frac{16-8}{16}-\frac{8}{20}-\frac{8}{80}=0$$

$$u_L(0^+)=8\times\frac{1}{2}=4\ \text{V}$$

$$\left.\frac{di_L}{dt}\right|_{t=0^+}=\frac{u_L(0^+)}{L}=4\ \text{A/s}$$



3.

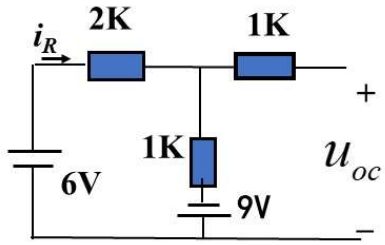


K 在 *t* =0时闭合. 求 i_R 并定性画出其波形.

解法一：

$$u_C(0^+)=u_C(0^-)=6\text{V}$$

0^+时刻以后的等效电路：



$$u_{oc}=1k\Omega\, i_R-9=-4V$$

$$R_{等}=1+\frac{2\times1}{2+1}=\frac{5}{3}\text{K}\Omega$$

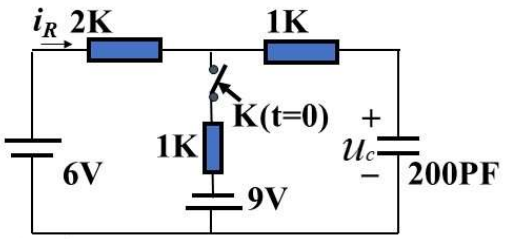
$$\tau=\frac{5}{3}\times10^3\times200\times10^{-12}=\frac{1}{3}\times10^{-6}\text{s}$$

$$u_C=-4+(6-(-4))\text{e}^{-\frac{t}{\tau}}\text{V}$$

$$i_C=C\frac{du_C}{dt}=-6\text{e}^{-\frac{t}{\tau}}\text{mA}\qquad i_R=\frac{6-(1k\Omega\, i_C+u_C)}{2k}=5-2\text{e}^{-\frac{t}{\tau}}\text{mA}\quad t\geq0^+$$



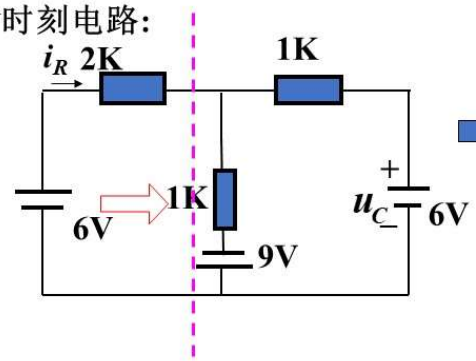
3.

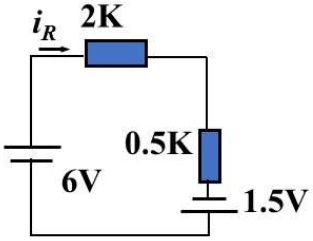


K 在 $t=0$时闭合. 求 i_R 并定性画出其波形.

解法二: $u_C(0^+)=u_C(0^-)=6\text{V}$

0^+时刻电路:



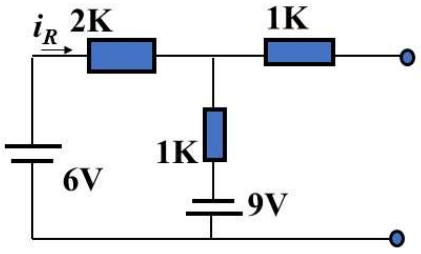


$$i_R(0^+)=\frac{6+1.5}{2.5}=3\text{mA}$$

$$i_R(0^+)=3\text{mA}$$

$t=\infty$时电路:

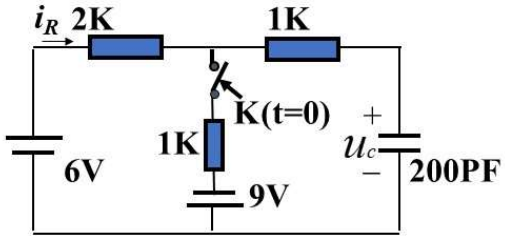
$$i_R(\infty)=\frac{6+9}{3}=5\text{mA}$$



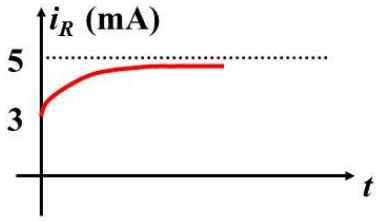
$$R_{等}=1+\frac{2\times1}{2+1}=\frac{5}{3}\text{K}\Omega$$

$$\tau=\frac{5}{3}\times10^3\times200\times10^{-12}=\frac{1}{3}\times10^{-6}\text{s}$$

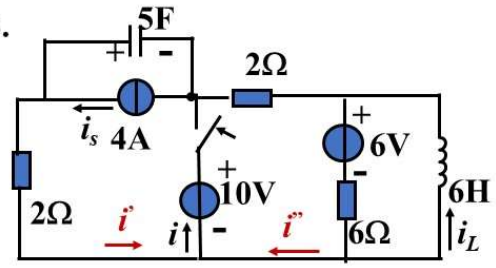
$$i_R=5+(3-5)\text{e}^{-\frac{t}{\tau}}=5-2\text{e}^{-\frac{t}{\tau}}\text{mA}\quad t\geq0^+$$



$$i_R = 5 + (3-5)e^{-\frac{t}{\tau}}\text{mA}$$



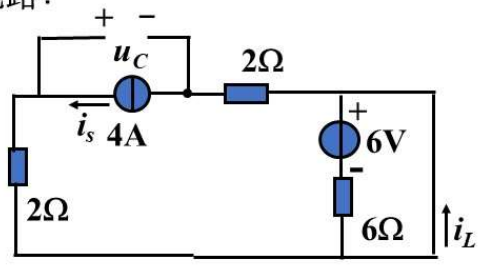
4.



开关在 $t=0$ 时刻闭合. 求 i.

$$i = i' + i''$$

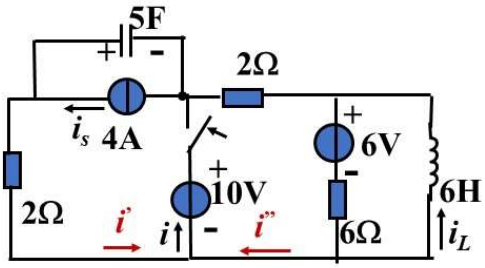
0^- 时刻电路:



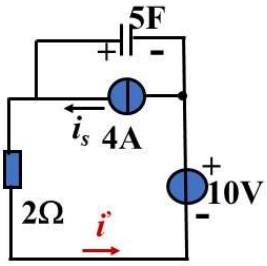
$$u_C(0^-) = 16\text{V}$$

$$i_L(0^-) = 3\text{A}$$

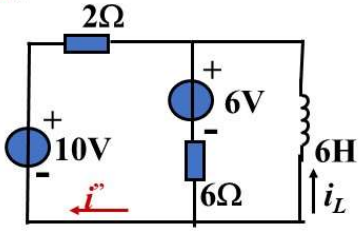
$$u_C(0^+) = 16\text{V} \qquad i_L(0^+) = 3\text{A}$$



求 i':

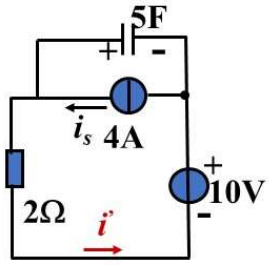


求 i'':



$u_C(0^+) = 16\text{V}$

$i_L(0^+) = 3\text{A}$



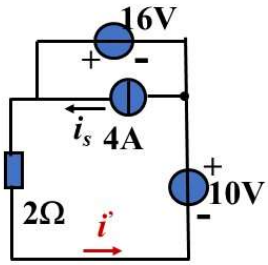
$u_C(0^+) = 16\text{V}$

$i'(0^+) = 13\text{A}$

$i'(\infty) = 4\text{A}$

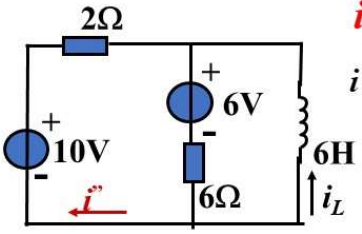
$\tau_1 = 10\text{s}$

$i' = 4 + 9e^{-0.1t}\,\text{A}$



$i_L(0^+) = 3\text{A}$

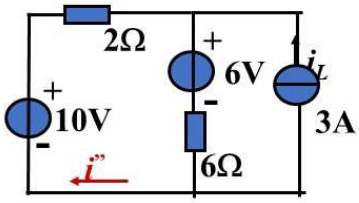
$$i''(0^+) = \frac{10-6}{2+6} - 3\times\frac{6}{2+6} = -1.75\,\text{A}$$



$i''(\infty) = 5\text{A}$

$\tau_2 = 4\text{s}$

$i'' = 5 - 6.75e^{-0.25t}\,\text{A}$



$i = i' + i'' = 9 + 9e^{-0.1t} - 6.75e^{-0.25t}\,\text{A} \quad t > 0$